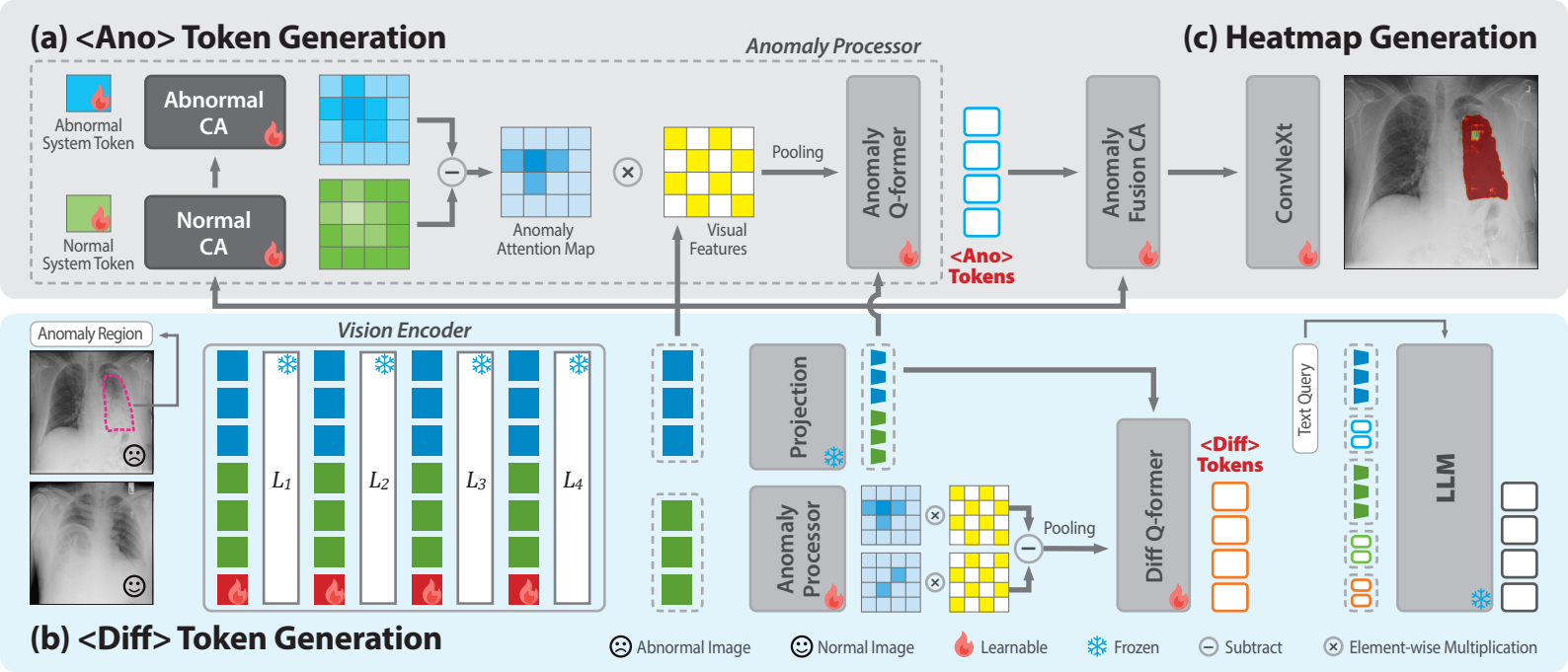
(a) <Ano> Token Generation
Anomaly Processor
(c) Heatmap Generation
Abnormal System Token
Abnormal CA
Normal System Token
Normal CA
Anomaly Attention Map
Visual Features
Pooling
Anomaly Q-former
<Ano> Tokens
Anomaly Fusion CA
ConvNeXt
Anomaly Region
Vision Encoder
L1
L2
L3
L4
Projection
Anomaly Processor
Pooling
Diff Q-former
<Diff> Tokens
Text Query
LLM
(b) <Diff> Token Generation
Abnormal Image
Normal Image
Learnable
Frozen
Subtract
Element-wise Multiplication